

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			burns and produces heat / acts as a fuel (1) forms carbon monoxide (1) carbon monoxide reduces the iron oxide / is a reducing agent (1) award last marking point for coke as a reducing agent if no reference to carbon monoxide	3			3		
	(b)	(i)		100g of CaCO ₃ can produce 56g of CaO (1) 8g CaCO ₃ can produce $\frac{56}{12.5}$ g of CaO or 4.48g of CaO (1) award (1) for any of following 4.48 to one decimal place is 4.5 therefore Charlie is correct 4.48 to one decimal place is 4.5 4.48 therefore Charlie is correct do not accept 4.5 and Charlie with no working alternative methods 0.08 mol CaCO ₃ (1) 0.08 × 56 = 4.48g CaO (1) award (1) for any of following 4.48 to one decimal place is 4.5 therefore Charlie is correct 4.48 to one decimal place is 4.5 4.48 therefore Charlie is correct			3	3	3	3

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				'trial and error' methods working backwards from any of the values given e.g. $\frac{4.4 \times 100}{56} = 7.86$ $\frac{4.5 \times 100}{56} = 8.04$ $\frac{5.0 \times 100}{56} = 8.93$ $\frac{100}{8} = 12.5$ $12.5 \times 4.4 = 55$ $12.5 \times 4.5 = 56.3$ $12.5 \times 5.0 = 62.5$ closest to 8.0g therefore Charlie closest to 56 therefore Charlie						
		(ii)		CaSiO ₃		1		1		
	(c)			4 Fe + 3O₂ → 2 Fe₂O₃ award (1) for formula award (1) for balancing only if formula is correct		2		2		
				Question 4 total	3	3	3	9	3	3